MOVIE THEATRE WEBSITE

ASSIGNMENT 3 TEST PLAN

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GitHub Assignment 3: https://github.com/omal0/cs250

TEST PLAN

This test plan describes the verification and validation activities for the Movie Theatre Website. It is intended to confirm that the system follows the Software Requirements Specification and the Software Design Specification. Testing covers the customer website, administrator website, application services, domain classes, repositories, database interactions, payment integration, and email notifications.

Testing Objectives and Scope

The main testing objectives are to verify that customers can browse movies, select showtimes, reserve seats, purchase tickets, and retrieve orders; verify that administrators can manage theatre information; prevent duplicate seat sales; correctly handle reservation expiration and payment failures; enforce authorization; and protect sensitive payment information.

Testing will include normal inputs, invalid inputs, boundary conditions, authorization failures, database conflicts, external-service failures, and security-related checks. Tests will be organized into unit testing, functional testing, and system testing.

Features and Classes Being Tested

The following components and features are included in the test scope:

- Movie and CatalogService: movie browsing, title searching, publishing, archiving, movie details, and showtime listings.
- Theatre and Showtime: theatre scheduling and prevention of overlapping showtimes, including cleaning time.
- Seat and SeatService: seat maps, seat availability, seat holds, and duplicate-seat prevention. ● SeatReservation: temporary holds, release operations, expiration, and conversion to sold tickets. ● TicketType and PriceSchedule: ticket prices, taxes, service fees, discounts, and total calculations. ● Order, Ticket, OrderService, and CheckoutService: order creation, confirmation, cancellation, retrieval, and ticket status.
- Payment, PaymentService, and PaymentProvider: approved, declined, failed, delayed, and unknown payment responses.
- UserAccount and AdministrationService: authentication, administrator authorization, and protected operations.
- EmailService and AuditLog: order notifications, payment-failure notices, cancellation notices, and administrative records.

Unit Testing Plan

Unit testing will test individual classes and methods independently. Mock repositories, payment providers, and email services may be used so that each unit can be tested without relying on the entire application. At least two unit test sets will be included in the Excel test case document.

Representative unit tests include:

- Test PriceSchedule.calculateTotal() with a valid subtotal, tax rate, service fee, and discount. The

expected result is an accurate final total.

- Test Showtime.hasConflict() with overlapping and non-overlapping showtimes, including the required cleaning period.
- Test SeatReservation.isActive() before and after the expiration time.
- Test UserAccount.isPermitted() with an administrator and a regular customer.
- Test Order.matchesRetrievalData() with matching and non-matching order numbers and email addresses.

Unit test inputs will include valid values, zero values, boundary values, expired timestamps, duplicate identifiers, invalid prices, and unauthorized user roles.

Functional Testing Plan

Functional testing will verify that multiple classes and services work together to complete a specific feature. These tests focus on behavior visible to customers and administrators. At least two functional test sets will be included in the Excel test case document.

Functional tests will include:

- Searching for a movie and displaying only valid published results.
- Selecting an available seat and creating a temporary reservation.
- Attempting to reserve a seat that has already been reserved or sold.
- Allowing a reservation to expire when checkout is not completed.
- Calculating the correct order total after selecting multiple ticket types.
- Processing an approved payment and converting reserved seats into sold tickets. ● Processing a declined payment without confirming the order.
- Retrieving a guest order using the correct order number and matching email address. ● Preventing a customer from accessing administrator functions.
- Sending an order-confirmation email after a successful purchase.

System Testing Plan

System testing will test the complete integrated Movie Theatre Website from the user interface through the application services, database, external payment provider, and email service. At least two system test sets will be included in the Excel test case document.

Complete system workflows will include:

- A customer browses a movie, selects a showtime, selects seats, chooses ticket types, submits payment, receives an order confirmation, and retrieves the order.
- Two customers attempt to purchase the same seat for the same showtime. The system must allow only one confirmed purchase.
- An administrator logs in, creates a movie, schedules a showtime, updates a ticket price, and views the resulting audit records.

- A customer begins checkout but does not complete payment. The seat reservation expires and becomes available again.
- A payment-provider failure occurs during checkout. The system must not create a falsely confirmed order.
- A user attempts to access administrator pages without administrator authorization. ● The system is inspected to verify that complete card numbers and security codes are not stored in the database, logs, audit records, or email messages.

Test Inputs and Test Vectors

The test cases will use the following inputs and test vectors:

- Published, unpublished, and archived movies; existing and non-existing movie titles. ● Valid, overlapping, adjacent, and cleaning-time-conflicting showtimes.
- Available, reserved, sold, inactive, and inaccessible seats.
- Active, expired, released, and converted seat reservations.
- Adult, child, and senior ticket types with valid and invalid prices.
- Tax rates, service fees, discounts, zero totals, and boundary monetary values.
- Approved, declined, timed-out, failed, and unknown payment responses.
- Valid and invalid order numbers, matching and non-matching guest email addresses. ● Registered customers, administrators, unauthorized users, and invalid login credentials. ● Missing, malformed, duplicate, and conflicting data.

Expected Results

A test is successful when the actual system behavior matches the expected result. The expected results are:

- Published movies appear in customer searches, while unpublished and archived movies do not appear as available movies.
- Conflicting showtimes are rejected, including conflicts caused by required cleaning time. ● Available seats can be held, while reserved or sold seats cannot be selected.
- Expired reservations release their seats for future customers.
- A seat cannot be sold more than once for the same showtime.
- Ticket totals, taxes, fees, discounts, and final charges are calculated accurately. ● Approved payments create confirmed orders and sold tickets.
- Failed or declined payments do not create confirmed orders.
- Guest orders require both the correct order number and matching email address. ● Only authorized administrators can use administrator functions.
- Order confirmations are sent only after successful payment.
- Sensitive card information is not stored or exposed.

Failure Cases

The test plan will specifically cover the following failure conditions:

- A customer searches for a movie that does not exist.
- An administrator attempts to schedule overlapping showtimes.
- A customer selects a seat that was already reserved by another customer.
- A reservation expires during checkout.
- Payment is declined, rejected, delayed, or returns an unknown result.
- An invalid ticket type or negative price is submitted.
- A guest enters an incorrect order number or email address.
- A regular customer attempts to access administrator functions.
- The database or external payment service becomes unavailable.
- An email notification fails after an order is confirmed.
- A request attempts to submit complete card details for storage.
- The system must respond with controlled error messages, preserve data consistency, avoid falsely confirming orders, and prevent sensitive information from being recorded.

Test Coverage

Unit tests provide coverage of individual methods and business rules. Functional tests provide coverage of connected features such as seat reservations, checkout, payments, order retrieval, and administrator operations. System tests provide coverage of complete customer and administrator workflows across all application layers.

Together, the tests cover the major domain classes, application services, external integrations, database operations, security rules, normal workflows, invalid inputs, and failure conditions identified in the Software Requirements Specification and Software Design Specification. Any failed test will be documented, corrected, and executed again to confirm that the defect has been resolved.